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RESEARCH ARTICLE

An Insight on the Timely Diagnosis of Diabetic Retinopathy Using Traditional and AI-Driven Approaches

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ABSTRACT Diabetic Retinopathy is a progressive microvascular complication of diabetes that requires early detection to improve patient outcomes. Traditional screening techniques, including fundus photography and optical coherence tomography, provide valuable diagnostic insights but have some limitations, like cost and technical complexity. Artificial intelligence is transforming the detection of diabetic retinopathy, moving away from traditional machine learning models that rely on manually created features to deep learning methods that allow for automatic feature extraction from retinal images. This systematic review investigates the evolution and diagnostic performance of AI-based techniques for DR detection. Studies were included if they applied machine learning or deep learning methods to retinal fundus or OCT images for DR classification. A total of 116 studies were included following comprehensive searches in databases such as PubMed, ScienceDirect, and IEEE Xplore, covering publications up to February 2024. Risk of bias was assessed in a representative sample of six studies, indicating a significant overall risk due to inadequate reporting on blinding and selective outcome reporting. Federated learning emerged as a promising alternative, enabling decentralized collaboration without compromising data privacy. Additionally, the growing focus on Explainable AI helps address the ''non-explainable'' nature of deep learning models by providing visual and textual explanations for predictions, thereby enhancing clinician trust and facilitating informed decision-making. By incorporating artificial intelligence with standard diagnostic frameworks, this research highlights the possibility for more accurate, scalable, and reachable diabetic retinopathy detection, paving the way for considerable advancements in ophthalmic problem management and enhancing patient care.

INDEX TERMS Diabetic retinopathy, ophthalmologists, convolutional neural network, retina fundus.

I. INTRODUCTION

Diabetes is a disease where blood glucose levels become extremely high due to the body's unfitness to produce or use insulin decently [1]. Insulin, a hormone free by the pancreas [2], allow glucose from carbohydrates to go in cells for energy. When this procedure is disrupted, glucose gather in the bloodstream, produce hyperglycemia [3]. Over period, high glucose amount harm blood vessels and nerves, grow the risk of difficulty like cardiovascular disease [4], nerve disablement, and eyesight loss. Diabetes mellitus, the technological term for this situation, differs from the infrequent diabetes insipidus [5], which act on fluid balance rather than glucose metabolism. Proper prescription and lifestyle improvement help manage diabetes successfully. Understanding diabetes is critical, not only because of its commonness but also because of its affect on overall health. By acknowledge its causes, outcomes, and management plans, individuals can take energetic steps to continue their well-being and prevent difficulties. Given the expand global bundle of diabetes and its similar problems, especially diabetic retinopathy, it is crucial to examine existing research

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to recognize growth in detection techniques. A structured review of latest studies can help recognize trends, challenges, and chances in this field. To attain this, a Systematic Literature Review (SLR) was manage to synthesize applicable research and provide a comprehensive overview of diabetic retinopathy detection techniques.

A. SYSTEMATIC LITERATURE REVIEW USING PRISMA

This section presents the SLR conducted to investigate the cutting-edge advancements in diabetic retinopathy detection. The study follows the PRISMA framework to establish a structured and reliable review process. Different databases such as Springer, IEEE, MDPI and ScienceDirect were inspected to identify superior, peer-reviewed studies. The inclusion and exclusion criteria and study selection process, are outlined below.

1) SEARCH STRATEGY

A comprehensive search was conducted to identify relevant studies on the diagnosis of diabetic retinopathy using traditional and AI-based methods. The following databases were included: PubMed, IEEE Xplore, ScienceDirect, Springer, MDPI, and others. The searches were carried out between March 5 and 20, covering studies published from January 2015 to February 2024. This search strategy used a combination of direct and indirect keywords related to diabetic retinopathy, artificial intelligence, machine learning, deep learning, federated learning (FL) and explainable AI (XAI). Only studies in English were included. Google Scholar were also used as a supplementary source to indetify literature or recent publications using words Diabetic Retinopathy stages, OCT, Fundus Images etc. Details of the search strategy for the databases are shown in the table1.

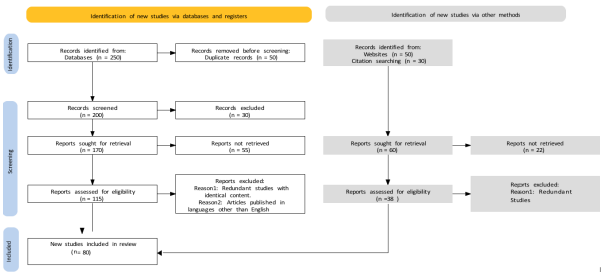


FIGURE 1. PRISMA flow diagram.

2) RESEARCH MECHANISM

The flowchart in Figure 1 tells the study selection process followed in this literature review. The identification stage stared with a total of 250 records retrieved from databases and an additional 80 records identified through websites and citation searching. After removing 50 duplicate records, a total of 200 records were screened for relevance. During the screening stage, 30 records were excluded due to irrelevance and 55 reports could not be retrieved. Then 165 reports are left for eligibility assessment from

TABLE 1. Search strategy summary.

Database	Search String Used	Date Searched	Year Range
PubMed	"Diabetic Retinopathy" AND ("Artificial Intelligence" OR "Deep Learning") AND (Detection OR Traditional Diagnosis)	5Mar 2025–7Mar 2025	2015–2025
IEEE Xplore	("DR stages" AND "Deep Learning") OR ("Federated Learning" OR "Explainable AI")	17Mar 2025–19Mar 2025	2015–2025
ScienceDirect	"Diabetic Retinopathy" AND ("Machine Learning" OR "Explainable AI" OR "Federated Learning")	17Mar 2025 – 19Mar 2025	2015–2025
Springer	"AI in Diabetic Retinopathy Detection" OR "XAI in Ophthalmology" OR "Federated Learning for DR Screening"	8Mar 2025–12Mar 2025	2015–2025
MDPI	"Diabetic Retinopathy" AND ("Deep Learning" OR "XAI" OR "Federated Learning")	12Mar 2025–16Mar 2025	2015–2025

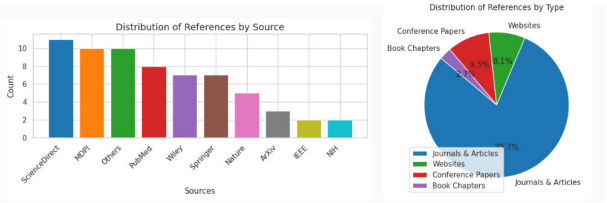


FIGURE 2. Relevant paper distribution based on data source and document type.

database searches and 38 reports from other sources. In the eligibility assessment stage, some reports were excluded for reasons such as redundancy (identical content) and language restrictions because we want only English language based articles. Ultimately, 80 studies were included in the final review, forming the basis of the literature synthesis. This approach ensured that only high-quality, relevant, and accessible studies contributed to the review, providing a robust foundation for further analysis. The given paper distribution is based on all the studies inlcuded, where are 80.The bar chart on the left presents the distribution of references by source, indicating that ScienceDirect, MDPI, and Others have the highest reference counts (10 each), followed closely by PubMed (8), Wiley (7), and Springer (7). Other sources like Nature, ResearchGate, IEEE, and ArXiv have fewer references, reflecting their lesser but still significant contribution. The pie chart on the right categorizes these sources into five main types: Journals (64.5%), Articles (15.8%), Websites (7.9%), Conference Papers (9.2%), and Book Chapters (2.6%). The dominance of journal references signifies a strong reliance on peer-reviewed and high-impact sources, ensuring academic rigor. Articles and conference papers add recent developments and emerging research,

while websites provide supplementary information. Book chapters contribute a small but valuable share, often offering foundational knowledge. This distribution ensures a balanced and well-rounded reference base for the research. After the eligibility assessment, 80 studies were included in the final review. However, 35 studies that seemed promising were excluded due to issues like non-English text, lack of AI focus, or weak methods.

3) PARADIGMS FOR INCLUSION AND EXCLUSION

A structured set of study protocols was developed to systematically select high-quality scholarly articles while excluding irrelevant or low-quality studies Table. 2. Three well-defined inclusion criteria were applied to ensure the selection of studies aligned with the research objectives. Additionally, exclusion factors were implemented to filter out studies lacking methodological rigor or relevance. This approach ensures that only credible and valuable research contributes to the literature review, maintaining academic integrity and reliability.

TABLE 2. Eligibility criteria for study selection.

Inclusion Criteria
Studies presenting AI-driven solutions for enhancing diabetic retinopathy detection.
Primary research papers rather than review or survey articles.
Articles that directly address at least one research question.
Scholarly publications published upto April 2025.
Exclusion Criteria
Articles published in languages other than English.
Redundant studies with identical content.

During the screening process, a total of 115 studies were reviewed for eligibility. After applying the defined inclusion and exclusion criteria, almost 35 studies were excluded based on factors such as language barriers, lack of relevance to diabetic retinopathy (DR), study type (e.g., reviews, meta-analyses, opinion pieces), or methodological limitations. Full-text Screening Articles that provide only partial insights into the keyword search are excluded if they fail to reference or establish a clear connection to a specific keyword listed in Table. 6.

TABLE 3. Excluded references based on criteria (Part 1).

Authors/References	Year	Reason for Exclusion
García, M., López, J. [6]	2021	Published in Spanish
Anton N, Doroftei B. [7]	2023	Review article
G. Latha, P. Aruna Priya [8]	2022	Focuses on glaucoma
Kawasaki, A., Kato, T. [9]	2022	Not about DR or AI
Wolfensberger TJ, Hamilton AM. [10]	2001	Published before 2014
Rahman, Rashda P. [11]	2024	Focuses on Keratoconus
Gert Litjens, Thijs Kooi [12]	2017	Redundant topic
Massin, Pascale [13]	2025	Published in French
Gao, Xian [14]	2024	Survey paper
Srinivasan, V., Krishnan [15]	2010	Sample size too small
Dong, Li, He [16]	2022	Not specific to DR
Naeem, Aleeza [17]	2025	No AI component
Sadda, S. R., Singh. [18]	2010	Lacks methodology

TABLE 4. Excluded references based on criteria (Part 2).

Authors/References	Year	Reason for Exclusion
Nevska AO, Pohosian OA. [19]	2024	Not peer-reviewed
American Academy of Ophthalmology [20]	2025	Published in Spanish
Deokate, Amol [21]	2025	No specific focus on DR
Shao, Yi, Jie [22]	2023	Redundant study
Md Mohaimenul Islam [23]	2020	Meta-analysis
Tan, Zachary [24]	2019	Opinion piece
Hatanaka, N., Murata [25]	2008	Before 2014
Guergueb, Takfarines [26]	2023	Focuses on glaucoma
Nkurunziza, Jean de Dieu [27]	2022	Not tied to RQs
De Fauw, Jeffrey, Ledsam [28]	2018	Lacks AI solutions
Sivaprasad, S., Gupta [29]	2024	Case study only
Ghanchi, Faruque [30]	2013	Clinical guideline

TABLE 5. Excluded references based on criteria (Part 3).

Authors/References	Year	Reason for Exclusion
Rohit Thanki [31]	2023	No clear focus on DR
Omar Ali, Wiem Abdelbaki [32]	2023	Systematic review
Maya, K V [33]	2018	Redundant study
Nagasaka, Y., Kato [34]	1987	Incomplete methodology
Gour, Neha [35]	2023	Too broad
Niall Patton, Tariq M. Aslam [36]	2006	Before 2014, no AI
Choudhary, Amit, Ahlawat [37]	2023	No validation/results
Grzybowski, Andrzej, Brona [38]	2020	Review article
Prathiba, V., Rema [39]	2011	No AI, not relevant
Carrera, Enrique V. [40]	2017	Flawed methodology

TABLE 6. Direct and indirect keywords used.

Core Keyword	**Diabetic Retinopathy**
Direct Keywords	Artificial Intelligence, Machine Learning, Deep Learning, Federated Learning, Explainable AI
Indirect Keywords	Ophthalmology, Fundus Images, DR Stages, OCT

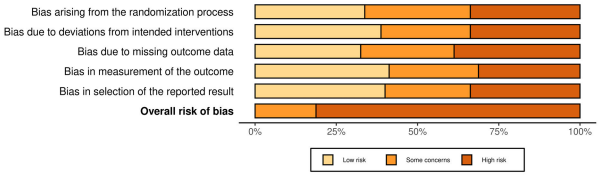


FIGURE 3. Risk of bias assessment summary.

4) RISK OF BIAS ASSESSMENT

Risk of bias was assessed for all 80 included studies using core domains adapted from the Cochrane Risk of Bias 2.0 tool [41], including randomization, blinding, handling incomplete outcome data, selective reporting and other potential sources of bias. Each study was independently rated across these domains and overall risk of bias judgment was assigned. Figure3 presents a summary of risk of bias assessments across the 80 included studies using ROBVIS tool [42].A significant portion of studies is rating as having high risk of bias. This elevated risk may, in part, reflect the heterogeneity of diagnostic methods included in the review. While several recent studies employed AI-based image analysis with standardized performance metrices, other relied on traditional clinical examination or manual grading of fundus images, which are often more subjective and prone to variations. Studies using traditional diagnostic approaches

were more likely yo lack detailed reporting, blinding or standardized outcome measures contributing to higher ROB scores.

5) DATASETS

This study leverages diverse data sets on diabetic retinopathy to enhance AI-driven detection. EyePACS, Kaggle DR, MESSIDOR, MESSIDOR-2, IDRiD, DIARETDB1, DRIVE, and STARE provide 2D fundus images, which support the detection of lesion, severity classification, and vessel segmentation. In contrast, OCTA includes 3D optical coherence tomography angiography scans, enabling detailed microvascular and structural analysis. These datasets, with varying resolutions and imaging conditions, serve as essential benchmarks for developing robust AI models in the research of diabetic retinopathy.

B. OVERVIEW OF DIABETES

Diabetes exists in multiple forms, each with unique characteristics. Pre-diabetes [43] is an early warning stage where blood sugar levels are elevated but not yet at Type 2 diabetes levels. Without intervention, it increases the risk of diabetes, heart disease, and stroke, although lifestyle changes can help prevent progression. Type 2 diabetes [44] is the result of insulin resistance [45], where the body struggles to use insulin effectively. Previously seen in adults, it is now rising among children due to poor lifestyle habits. Type 1 diabetes [46], an autoimmune disorder, destroys insulin-producing cells, requiring lifelong insulin therapy. Though its exact cause is unclear, it typically develops in childhood or early adulthood. Gestational diabetes [47] occurs during pregnancy due to hormonal shifts affecting insulin function. While it usually resolves after childbirth, it raises the risk of developing Type 2 diabetes later. Recognizing diabetes symptoms is crucial for early diagnosis. Frequent urination(polyuria) occurs as the body expels excess glucose, leading to excessive thirst (polydipsia) due to dehydration. Unexplained weight loss happens when the body breaks down fat and muscle for energy, causing intense hunger and fatigue. Blurred vision, slow-healing wounds, and frequent infections arise as high blood sugar weakens circulation and immune response. Nerve damage (neuropathy) may cause tingling or numbness in the hands and feet. Type 1 diabetes progresses rapidly, often within weeks, sometimes leading to diabetic ketoacidosis(DKA) [48], a severe condition causing nausea, vomiting, confusion, and fruity-smelling breath. Type 2 diabetes, however, develops gradually and can go unnoticed for years. Some individuals exhibit darkened skin patches, a sign of insulin resistance [45].Gestational diabetes [47] is usually detected during routine screening, though some women may experience mild symptoms like thirst and frequent urination. Understanding the causes of diabetes is key to prevention. Type 2 diabetes stems from insulin resistance, worsened by obesity, inactivity, poor diet, genetics, and hormonal imbalances. The pancreas compensates by producing more insulin, but over time, it fails, causing high blood sugar. In Type 1 diabetes and latent autoimmune diabetes in adults (LADA) [49], the immune system mistakenly destroys insulin-producing cells, leading to insulin deficiency. Hormonal imbalances also play a role. Gestational diabetes arises when pregnancy hormones induce insulin resistance [47]. Disorders like Cushing's syndrome and acromegaly [50] similarly increase diabetes risk. In some cases, pancreatic damage from disease, surgery, or injury leads to Type 3c diabetes. Additionally, genetic mutations cause rare forms such as maturity-onset diabetes of the young (MODY) [51] and neonatal diabetes, both affecting insulin production from birth. Regardless of type, diabetes disrupts blood sugar regulation, leading to severe health risks if unmanaged. Early detection, lifestyle adjustments, and proper treatment are key to effective management [52].

Diabetes leads to severe complications, categorized as microvascular and macrovascular. Microvascular issues [53] like neuropathy, nephropathy, and retinopathy result from prolonged high blood sugar damaging small blood vessels, while macrovascular complications such as cardiovascular disease, stroke, and PAD increase life-threatening risks. A major concern is diabetic foot syndrome [54], caused by neuropathy, PAD, and infections, often leading to ulcers and amputations. Other effects include vision impairment (retinopathy), gastroparesis, extreme thirst, dry skin, infections, and stroke risk Fig. 4. Among these, diabetic retinopathy is particularly alarming, as it progressively damages vision and may cause blindness. The next section explores its types, effects, and impact.

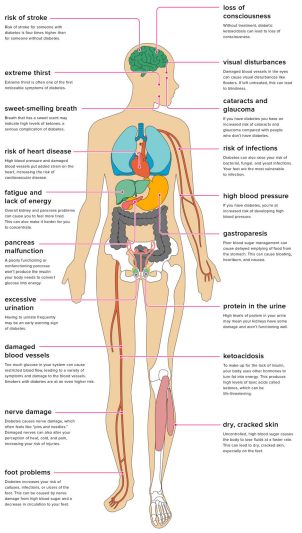


FIGURE 4. Effects of diabetes [55].

If left unmanaged, diabetes can cause severe complications, with diabetic retinopathy [56] being one of the most alarming. This progressive microvascular disorder silently damages retinal blood vessels, often going unnoticed until vision loss or blindness occurs. It is the leading cause of vision impairment among working-age adults in the Western

world. Who is at risk? Anyone with diabetes—diagnosed or not—can develop diabetic retinopathy. The risk increases with age and the duration of diabetes, but poor blood sugar control and unstable blood pressure accelerate the damage, making early management critical. Fig. compares a normal retina with one affected by diabetic retinopathy, highlighting severe damage like hemorrhages, aneurysms, abnormal vessel growth, hard exudates, and ''cotton wool'' spots. These changes, driven by prolonged high blood sugar, can lead to blurred vision or blindness if left untreated.

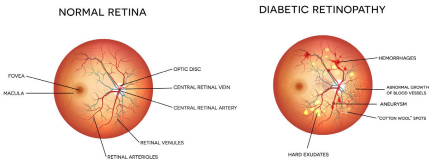


FIGURE 5. Normal retina vs diabetic retinopathy [57].

Diabetic Retinopathy (DR) is classified into non-proliferative (NPDR) and proliferative (PDR) stages [58]. NPDR involves microaneurysms, hemorrhages, intraretinal microvascular abnormalities (IRMA), and cotton wool spots, progressing through mild, moderate, severe, and even the very severe stages, illustrated in Fig. 6 [58]. Mild NPDR presents with microaneurysms and hemorrhages in all quadrants, while moderate NPDR shows increased lesions, venous beading, and IRMA in up to three quadrants. Severe NPDR is diagnosed with extensive hemorrhages, venous beading in at least two quadrants, or severe IRMA in one quadrant. Very severe NPDR meets at least two severe NPDR criteria without PDR signs [58].

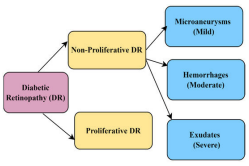


FIGURE 6. Stages of DR [59].

Diabetic Retinopathy carries multiple risks, some unavoidable, but others—such as hypertension, obesity, dyslipidemia, poor glycemic control, and nephropathy—can be managed through lifestyle changes and medical intervention [60], [61]. Early DR often develops silently, with occasional vision disturbances like difficulty focusing. As high blood sugar weakens retinal blood vessels, they rupture, leaking blood into the vitreous humor [62], causing floaters or streaks. While these may fade, untreated bleeding can lead to scar tissue formation and vision distortion. Diabetes-induced vascular damage triggers abnormal vessel growth, increasing the risk of further bleeding and blindness. Timely intervention is crucial, but traditional diagnostic methods like fundus imaging and fluorescein angiography require expertise and may be inaccessible in resource-limited settings. Advances in AI and deep learning now offer automated DR detection, improving diagnostic accuracy and accessibility. The following sections explore both traditional and AI-based approaches, assessing their strengths and limitations.

II. DIAGNOSIS TECHNIQUES OF DIABETIC RETINOPATHY DETECTION

The detection of diabetic retinopathy has transitioned from conventional methods to advanced AI techniques, enhancing accuracy and efficiency. This section explores traditional approaches, followed by ML, DL, FL, and XAI-based methods.

A. TRADITIONAL DIAGNOSIS TECHNIQUES

Medical advancements continue to enhance diabetic retinopathy diagnosis, with retinal imaging [63] playing a crucial role in detecting abnormalities. While traditional imaging techniques are essential, they come with challenges like high costs, specialized equipment, and reliance on expert interpretation, limiting large-scale screening. Additionally, 2D imaging struggles to capture depth-related conditions like diabetic macular edema (DME) [64], and manual assessments risk subjectivity in diagnosis. Among these methods, fundus photography stands out for its high-resolution imaging of the retina, optic disc, and macula, making it a key tool in detecting microaneurysms, hemorrhages, exudates, and neovascularization in DR. The Early Treatment Diabetic Retinopathy Study (ETDRS) established stereoscopic color fundus photography [65] as the gold standard, but it remains time-consuming and resource-intensive. To improve efficiency, alternative imaging techniques have been explored. Single-field fundus photography has a sensitivity of 78% and specificity of 86% [66], while two-field imaging improves sensitivity to 96% and specificity to 89% [67]. Three-field imaging further enhances accuracy, achieving 92% sensitivity and 97% specificity [68], making two- or three-field imaging the preferred choice for balanced accuracy and patient comfort. Mydriatic fundus photography enhances image quality by pupil dilation, improving sensitivity to 97% for vision-threatening DR (VTDR) [69]. However, concerns about acute angle-closure glaucoma from dilation persist [70]. As a result, nonmydriatic fundus photography Fig.7 has gained traction as a patient-friendly alternative, despite its higher technical failure rate due to media opacities, small pupils, and stereoscopic limitations. Despite these drawbacks, studies show nonmydriatic imaging can achieve up to 98% sensitivity and 90% specificity for VTDR detection [69], making it a valuable tool in primary care screening. **Optical Coherence Tomography (OCT)** has revolutionized retinal imaging, providing high-resolution, cross-sectional views of retinal structures. Often referred to as an ''optical ultrasound,'' OCT surpasses traditional methods by enabling real-time, non-invasive visualization of retinal layers, making it indispensable for diagnosing diabetic retinopathy (DR), diabetic macular edema (DME), and glaucoma. Unlike fundus photography, which struggles with DME detection [64], OCT delivers 2D and 3D



FIGURE 7. A nonmydriatic digital retinal photograph of a patient with severe nonproliferative diabetic retinopathy displays dot, blot, and flame hemorrhages along with hard exudates. This image was captured using the TRC-NW8 device (Topcon Medical Systems, Oakland, NJ). [71].

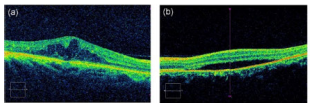


FIGURE 8. Optical coherence tomography (OCT) imaging effectively examines structural irregularities present in diabetic macular edema (DME), such as (A) cystic alterations within the retina and (B) fluid buildup beneath the retinal layers. [71].

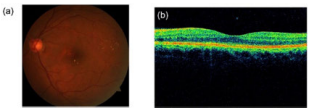


FIGURE 9. Optical coherence tomography (OCT) imaging effectively examines structural irregularities present in diabetic macular edema (DME), such as (A) cystic alterations within the retina and (B) fluid buildup beneath the retinal layers. [71].

imaging, capturing structural abnormalities with ultra-fast scanning [72].OCT utilizes low-coherence interferometry to perform an optical biopsy, allowing precise measurement of retinal thickness, subretinal fluid, and hemorrhages, aiding in early disease detection and treatment monitoring [73]. Its ability to conduct quantitative retinal analysis enhances DR and DME management, ensuring timely therapeutic interventions [74]. Studies show that subclinical DME patients face a 27% risk of significant retinal thickening within a year, rising to 38% in two years [75], reinforcing OCT's predictive capability. The advancement of Spectral-Domain OCT (SD-OCT) [76] has further improved DME assessment, surpassing conventional central subfield thickness measurements. This next-generation imaging detects intraretinal cysts, subretinal fluid, and vitreo-retinal traction, offering precise pathophysiological insights Fig.6. A Cochrane systematic review highlights OCT's 78% sensitivity and 86% specificity in DME detection, demonstrating its superiority over fundus biomicroscopy and stereophotography. These advancements establish OCT as a cornerstone of modern ophthalmology, enabling personalized, proactive eye care. As OCT becomes a cornerstone in modern eye care, it is now a key diagnostic tool for DME [77]. However, its role in DR screenings alongside fundus imaging remains debated, as the correlation between retinal thickening on OCT and fundus photography findings is moderate. OCT is particularly valuable for confirming suspected DME cases detected on fundus images, reducing false positives and unnecessary specialist referrals as shown in Figure. 8.In the UK, an SD-OCT-based DR screening system significantly cut down hospital referrals, with Dodson's study revealing that only 26% of DME cases flagged by fundus imaging exhibited macular swelling on SD-OCT. Beyond screening, OCT plays a crucial role in monitoring DR progression, offering histological accuracy in detecting microvascular irregularities and neovascularization [78]. These capabilities reinforce OCT's growing significance in precision-driven retinal disease management. **Scanning laser ophthalmoscopy (SLO)** is an advanced imaging technique that uses laser light to illuminate the retina, replacing traditional bright white light flashes. High-quality images are essential for accurate diabetic retinopathy assessment, which typically requires pupil dilation in standard fundus photography. Confocal SLO scans the retina point-by-point with a focused laser beam, capturing reflected light through a small aperture called a confocal pinhole. This blocks light from outside the focal plane, producing sharp, high-contrast images without pupil dilation.cSLO offers advantages over traditional fundus photography, including improved patient comfort due to lower light intensity, video capabilities, and effective imaging for patients with poor pupil dilation. Diabetic patients, who often struggle with pupil dilation and make up a large portion of those with vision issues [79], benefit significantly from cSLO, making it a valuable tool for DR screening. Standard fundus photography struggles to image the peripheral retina, where DR lesions may be missed. Ultra-widefield (UWF) imaging addresses this by combining laser light and confocal laser scanning microscopy to produce high-resolution images. By integrating SLO with an ellipsoidal mirror, UWF technology captures a single image covering up to 200Â° as shown in Figure. 10, compared to the 45Â°–50Â° range of standard fundus photography. Devices utilizing UWF imaging include the Optos retinal imaging systems (e.g., Optomap 200Tx, Daytona, and California Imaging Systems by Optos, Marlborough, MA) and the Heidelberg HRA (Heidelberg Engineering, Carlsbad, CA). The Heidelberg HRA includes a noncontact UWF module that easily attaches to Spectralis or HRA2 camera heads, enhancing its imaging capabilities.

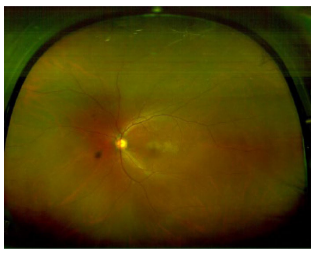


FIGURE 10. An ultra-widefield (UWF) image from the Optos P200MA (Optos, Marlborough, MA) offers a broad retinal view, enhancing peripheral visualization. [71].

TABLE 7. Evaluation of imaging techniques for diabetic retinopathy detection.

Technique	Key Features	Challenges	Sensitivity /Specificity
Fundus Photography	Standard DR imaging.	Time-consuming, expert-dependent.	Varies (ETDRS standard).
Single-Field Fundus	Quick screening.	Lower accuracy.	78% / 86% [66]
Two-Field Fundus	Improved accuracy.	More imaging time.	96% / 89% [67]
Three-Field Fundus	Balanced accuracy.	Resource-intensive.	92% / 97% [68]
Mydriatic Fundus	High-quality images.	Risk of glaucoma.	97% (VTDR) [69]
Nonmydriatic Fundus	No dilation, patient-friendly.	High failure rate.	98% / 90% [69]
OCT	High-resolution 2D/3D imaging.	Expensive, moderate correlation.	78% / 86% (DME) [76]
SD-OCT	Advanced structural analysis.	High cost, expertise needed.	Reduces false positives.
SLO	High-contrast imaging, no dilation.	Costly, requires expertise.	Improved DR detection [79]
UWF Imaging	200° retinal view, detects peripheral DR.	Expensive, limited availability.	Higher lesion detection vs. standard fundus

B. ADVANCEMENTS IN AI-DRIVEN DIABETIC RETINOPATHY DIAGNOSIS

Traditional methods have limitations in scalability and accuracy. AI-driven techniques, particularly machine learning, offer automated, data-driven solutions for enhanced DR detection.

1) MACHINE LEARNING APPROACHES

Machine Learning, a subset of AI, has revolutionized diabetic retinopathy diagnosis by enabling automated pattern recognition and predictive analysis [80]. Traditional DR detection still heavily relies on manual assessments, making scalability and consistency challenging. ML-based models address these limitations by analyzing vast retinal datasets with high precision and efficiency. Among the various ML techniques, Support Vector Machines, Decision Trees, Random Forest, K-Nearest Neighbors, Logistic Regression, and Hybrid ML approaches play a crucial role in ensuring robust and scalable diagnostic solutions.**Support Vector Machines** are machine learning models used for object classification by identifying an optimal decision boundary. [80] Several studies have utilized SVMs for automated diabetic retinopathy detection using retinal images. Carrera and González [40] proposed a computer-assisted system for detecting NPDR using digital image processing. Their study employed the Messidor Database [81], consisting of 1200 eye fundus images captured via non-mydriatic retinograph at varying resolutions (1400 × 960, 2240 × 1488, 2304 × 1536 pixels). Images were categorized into four NPDR severity levels: Normal, Mild, Moderate, and Severe NPDR. For experiments, 400 images were labeled on a 4-grade NPDR scale, while 301 images (152 normal, 149 severe) were used for binary classification.

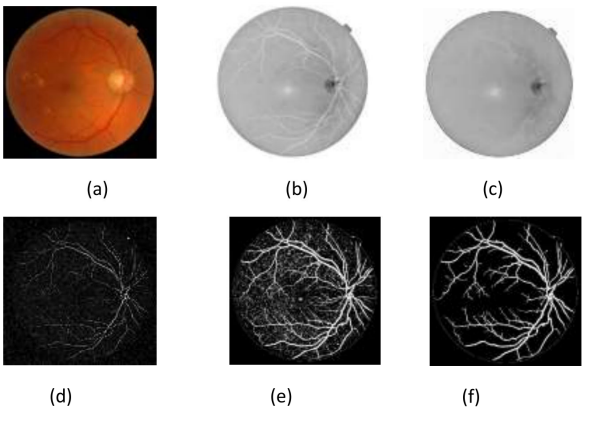


FIGURE 11. Blood vessel detection [81].

Fig.11:(a) Original image: Unprocessed blood vessel image from medical imaging.(b) Magenta component: Magenta channel extracted to highlight vessels.(c) Morphological processed image: Enhanced vessel structure via morphological operations.(d) Difference between b and c: Pixel differences between magenta and morphological images.(e) Binarized image: Black-and-white thresholded image of vessels.(f) Noise-reduced image: Binarized image with noise filtered out.

Blood Vessel Segmentation The RGB image is converted to CMY color space, extracting the magenta component. Morphological operations (erosion, opening, dilation) remove noise. The difference between the magenta component and the processed image is binarized to compute blood vessel density. Microaneurysm Detection The green channel is extracted, and blood vessels are hidden using a background mask. Edge detection and hole-filling algorithms identify microaneurysms, followed by filtering based on size and shape.

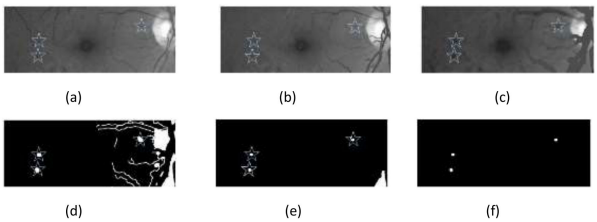


FIGURE 12. Microaneurysm detection [81].

Fig.12: (a) Green Component, [81](b) Vessel Masking, (c) Dilation, (d) Edge Detection, (e) Pre-selected, (f) Final Microaneurysms.

Hard Exudates Detection The magenta component is extracted, binarized, and the optic disc is removed using Hough transform. The final image is analyzed for hard exudate density.

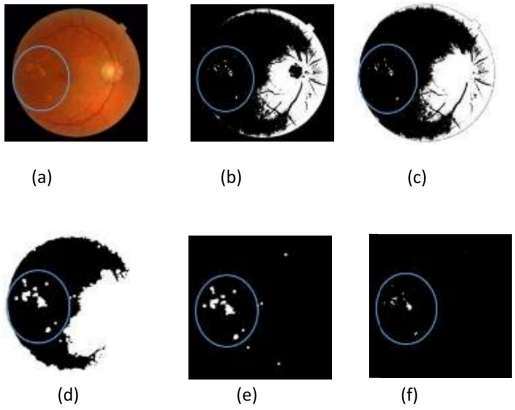


FIGURE 13. Hard exudates detection [81].

Fig.13: (a) Original Image, [81](b) Color Segmentation, (c) Background Removal, (d) Dilation, (e) Pre-selected, (f) Final Hard Exudates. After preprocessing, eight quantitative features were extracted: 1) Standard deviation of RGB components, 2) Blood vessel density, 3) Possible and actual microaneurysm count, 4) Hard exudate density, 5) Green component entropy. For classification, SVM with a linear kernel achieved optimal results. A one-vs-one SVM classified NPDR into severity grades, while a decision tree used the Gini index with a max depth of 4. Binary classification:SVM achieved 92.4% accuracy, 87.3% sensitivity, and 97.4% specificity, outperforming the decision tree. Multi-class (4-grade NPDR): SVM reached 85% accuracy, with blood vessel density, actual microaneurysm count, and blue component standard deviation as key features [40]. A notable advancement in SVM is the study by Mona Leeza, published in 2019, titled ''Detection of Severity Level of Diabetic Retinopathy Using Bag of Features Model''. This research introduced a Bag of Features (BoF) model combined with a radial basis function (RBF) kernel SVM to classify retinal images into five DR severity levels. The methodology involved extracting Speeded-Up Robust Features (SURF) and Histogram of Oriented Gradients (HOG) features from retinal images, enabling the model to identify critical patterns indicative of DR severity. The study utilized the EyePACS dataset, a widely recognized dataset containing retinal images used for DR classification, ensuring a diverse and representative sample for training and evaluation. The model achieved a sensitivity of 95.92% and a specificity of 98.90%, highlighting the robustness of SVM when integrated with advanced feature extraction Techniques [82]. By Continuing the research of SVM applications in diabetic retinopathy detection, more latest studies have evaluated the effectiveness of SVM models. The survey by Nissen et al. in 2023, investigates the success of a SVM based tool in detecting diabetic retinopathy (DR) within a clinical setting. The researchers trained an SVM model retinal images to check the difference between DR and non-DR cases. The dataset used in this study consists of 1,001 retinal images from 1,001 patients who had participated in the Danish National Screening Programme, verifying the applicability of model to real-world problems. This model gives the sensitivity of 84.9%, specificity of 92.0%, positive predictive value of 84.9%, and negative predictive value of 92.0%, describing the robust performance of SVM in DR detection [83]. One more enhancement in DR prediction is the study by P. Anitha and P. R. Tamilselvi in 2024. The title of the above mentioned study is ''Diabetic Retinopathy Prediction Using Soft Computing-Based Fuzzy-SVM Integrated Diabetic Retinopathy Prediction Framework (SC-FSIDR-PF).'' This research introduces the SC-FSIDR-PF, which combines fuzzy logic with SVM techniques to get the better accuracy. The framework utilizes soft computing to handle the fundamental uncertainties in medical data and improving the validity of the model. In this study the Indian Diabetic Retinopathy Image Dataset is used which consists of retinal images and associated clinical information from patients, ensuring a comprehensive analysis. The given framework accomplish an accuracy of 98%, 96% sensitivity, and 97% specificity, demonstrating its success in accurately predicting DR [84]. SVM-based models have demonstrated high effectiveness in diagnosing diabetic retinopathy, achieving strong accuracy and robustness. However, their dependence on feature selection and kernel tuning can be computationally intensive. Random Forest, an ensemble learning technique, provides enhanced generalization and scalability to address these challenges. The following section examines its role in DR detection.

Random Forest is a supervised learning technique that constructs multiple decision trees using the bagging approach to enhance predictive accuracy and stability. Key hyper-parameters include *n-estimators* (number of trees), *max-features* (features per split), *min-samples-leaf* (minimum samples per leaf), and *n-jobs* (CPU utilization). Additionally, *oob-score* provides built-in cross-validation [85]. Recent advancements integrate Random Forest with deep feature extraction to improve diabetic retinopathy detection. Traditional models rely on handcrafted features, limiting generalizability. To address this, a hybrid approach using ResNet-50 for feature extraction followed by Random Forest classification was proposed. The study utilized *EyePACS* and *Messidor-2* datasets [86], applying preprocessing steps such as resizing, normalization, adaptive histogram equalization for contrast enhancement, and Gaussian filtering for noise reduction. Deep feature extraction was performed using a pretrained ResNet-50 model [87], leveraging transfer learning to extract meaningful patterns. The fully connected

layers were removed, and deep features from the last convolutional layer served as input for the Random Forest classifier. Hyperparameter tuning optimized accuracy and computational efficiency The Hybrid ResNet-50 + Random Forest approach achieved 92.6% accuracy, 91.2% sensitivity, and 94.3% specificity on the EyePACS and Messidor-2 datasets, surpassing traditional methods (85-88% accuracy) [88]. While this approach automates feature extraction and improves accuracy, fully deep learning-based models, such as end-to-end CNNs, integrate feature extraction and classification, potentially enhancing diagnostic performance further. A highly effective diagnostic model is not just about

TABLE 8. Comparison of machine learning techniques for diabetic retinopathy detection.

ML Technique	Accuracy (%)	Sensitivity (%)	Specificity (%)
SVM (Linear Kernel) [40]	92.4	87.3	97.4
SVM (One-vs-One for NPDR) [40]	85.0	-	-
SVM (Bag of Features + RBF Kernel) [82]	95.92	95.92	98.90
SVM (Clinical Study) [83]	-	84.9	92.0
Fuzzy-SVM (SC-FSIDR-PF) [84]	98.0	96.0	97.0
Hybrid ResNet-50 + Random Forest [88]	92.6	91.2	94.3

making predictions—it's about making the right predictions. Accuracy serves as the foundation, reflecting how well the system distinguishes between healthy and affected individuals. However, in medical diagnostics, accuracy alone is not enough. A model must be sensitive, ensuring that no true cases of diabetic retinopathy go undetected. Missing a single patient could mean irreversible vision loss, making sensitivity (recall) a critical metric for early and effective intervention. The model must be specific, avoiding unnecessary medical tests for those people who are disease-free. Specificity ensures that the people who are only truly affected with disease receive further testing, avoiding false alarms. A well-balanced system seamlessly integrates these metrics, offering reliable, precise and clinically reliable results - lays the foundation for AI-driven advances in the detection of DR.

2) DEEP LEARNING

Deep learning, is a subset of machine learning that transform disease detection by enabling automated pattern recognition and feature extraction from datasets, thus minimizing the dependence on human involvement [89]. In diabetic retinopathy detection, deep learning significantly enhances diagnostic accuracy, minimizing reliance on manual feature selection. By leveraging supervised artificial neural networks, retinal images can be processed with remarkable precision, achieving detection accuracy as high as 97.13% [90]. Given that DR progressively affects the retina and may lead to vision impairment, early detection through computer-aided diagnosis (CAD) systems becomes critical in preventing irreversible damage. These systems integrate artificial intelligence and image processing techniques, significantly reducing dependence on manual diagnosis while improving consistency [91]. Moreover, CAD systems have been widely adopted in detecting other complex medical conditions, such as brain tumors and glaucoma, providing valuable assistance to healthcare professionals [92]. Remarkably, these systems demonstrate diagnostic performance comparable to human experts, contingent on the quality of the feature extraction methodologies employed [93]. Deep learning—particularly convolutional neural networks—has emerged as a transformative technology in medical image analysis [94]. CNN-based approaches have been extensively applied in DR diagnosis, demonstrating their efficacy in extracting retinal features such as blood vessels, exudates, and hemorrhages with high precision. For instance, Nayak et al. proposed a CNN model capable of classifying retinal images into non-DR, NPDR, and PDR categories, achieving an impressive classification accuracy of 93%, with a sensitivity of 90% and specificity of 100% [95]. Similarly, Pratt et al. leveraged CNNs with data augmentation techniques to enhance the detection of retinal abnormalities, including hemorrhages and microaneurysms, achieving 75% accuracy, 30% sensitivity, and 95% specificity on a Kaggle dataset containing 80,000 images [96]. Furthermore, Shaban et al. developed a CNN-based classifier trained on 101 fundus images, successfully categorizing DR into four severity stages: non-DR, NPDR, severe NPDR, and PDR [97]. Figure 14 illustrates how

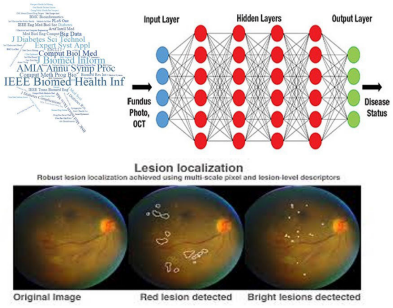


FIGURE 14. Hierarchical learning mechanism in deep learning models [91].

deep learning models process medical data through sequential layers, refining extracted information into hierarchical representations. The input layer extracts raw image pixels, which are subsequently analyzed by convolutional layers that detect fundamental patterns, such as edges and textures. The segmentation layer further organizes these patterns into meaningful structures, enabling the final classification layer to diagnose diseases like DR with minimal human intervention. This hierarchical learning approach eliminates the need for manual feature selection while optimizing model performance through backpropagation, mimicking the human visual cortex. Consequently, deep learning-based methodologies facilitate scalable, automated DR screening, enhancing efficiency in clinical settings [91]. CNNs, inspired

by the human visual system, excel in image classification tasks due to their ability to learn spatial hierarchies of features [98]. Their architecture consists of convolutional layers for detecting localized patterns, pooling layers for dimensionality reduction, and fully connected layers for final classification [99]. Training involves minimizing a loss function:

$$L(w) = \frac{1}{n}\sum_{i=1}^{n} l(z_i, f(x_i; w)), \tag{1}$$

where n represents the sample size, z_i the ground truth label, and $f(x_i; w)$ the predicted outcome. Optimization techniques such as backpropagation and gradient descent iteratively refine the model's parameters [100]. Xu [93] validated CNN-based approaches using a dataset of 800 training and 200 evaluation images ($224 \times 224 \times 3$), demonstrating the model's proficiency in medical imaging applications.

The advancements in CNN-based DR detection have been remarkable. Gulshan et al. [101] employed the Inception-v3 model on 128,175 retinal images, attaining 90.3% sensitivity and 98.1% specificity on the EyePACS-1 dataset. However, challenges such as dataset limitations and grading inconsistencies necessitate augmentation strategies and expert validation. To improve interpretability, Quellec et al. [102] integrated heatmaps, generalizing backpropagation to pinpoint lesion-specific contributions, achieving robust results on over 200,000 images. Yang et al. [103] introduced a two-stage deep CNN framework for lesion localization and DR severity grading, utilizing an imbalanced weighting strategy to improve classification performance on 23,595 Kaggle images, achieving results comparable to ophthalmologists. A [104] VGG16 transfer-learning network on KUB X-rays reached 97.4 % test accuracy for stone detection. Layer-wise relevance propagation then highlighted the exact image regions driving each prediction. This approach shows that deep CNNs can reliably flag kidney stones. Integrating such a tool into routine scans could speed diagnosis and improve patient outcomes. Additionally, Brahami et al. [105] developed a hybrid model integrating VGG16 and VGG19, attaining 90.6% accuracy and 95% recall on a dataset of 5,584 images, further enhancing clinical applicability. Collectively, these advancements signify a paradigm shift in DR diagnosis, transitioning from traditional image classification to interpretable, lesion-aware AI systems. While current CNN models achieve near-human diagnostic performance, real-world clinical validation remains essential for widespread adoption [87]. A ResNet-18 network applied to a publicly available Kaggle set hit 99.9 % accuracy in training and 96.7 % on unseen images. This result confirms that convolutional models can reliably flag diabetic retinal changes. Embedding such a tool into routine screenings could speed exams, boost care quality, and cut diabetes-related vision loss. These strides point toward more transparent, lesion-focused AI in DR diagnosis, though real-world trials remain crucial before wide use. Binary classification of DR using deep learning has evolved substantially, progressing from basic CNN models employing rudimentary augmentation techniques to sophisticated ensemble architectures. Early CNN models achieved 94.5% accuracy in distinguishing referable from non-referable DR cases [106]. However, these models often lacked interpretability, emphasizing raw accuracy over clinically meaningful feature extraction. The integration of pretrained architectures such as ResNet34 and VGG-16, coupled with preprocessing techniques like histogram equalization and noise reduction, significantly enhanced robustness. ResNet-based models achieved 85% accuracy, while VGG-16 architectures with dropout regularization attained an AUC of 98.2%, improving diagnostic reliability [106]. Recent advancements have focused on ensemble learning, leveraging models like Inception V3, Inception-ResNet-V2, and ResNet152 in conjunction with Adaboost optimization to harness their complementary strengths. By combining different deep learning techniques, the team achieved an accuracy of 88.21%, which really highlights the benefit of using a hybrid approach. Moreover, lesion-centric methodologies, such as weighted-path CNNs (WP-CNNs), prioritize critical features like microaneurysms and hemorrhages, achieving 94.23% accuracy, with red lesion detection models demonstrating a sensitivity of 0.94 and an AUC of 0.912 [106]. Despite these advancements, binary classification approaches struggle to address DR severity grading, which is crucial for treatment planning. While segmentation models attain high accuracy on vessel detection tasks—97.67% on DRIVE and 98.13% on STARE [107]—multi-class classification remains challenging due to imbalanced datasets and overlapping pathological features. To overcome these limitations, federated learning (FL) frameworks such as DRFL [108] have emerged, facilitating collaborative training across institutions without compromising data privacy. By integrating Federated Averaging (FedAvg) with multi-scale feature extraction, DRFL achieved 98.6% accuracy and 99.3% specificity on MESSIDOR-2 and IDRiD datasets, demonstrating its potential for large-scale deployment. However, real-world validation and seamless integration into clinical workflows remain imperative to bridge the gap between technological innovation and practical application, ensuring AI-driven DR detection meets the stringent requirements of ophthalmologists and healthcare providers.

3) FEDERATED LEARNING (FL)

Advanced deep learning systems require large datasets, but healthcare privacy laws fragment patient data into isolated institutional ''islands'' [113], risking overfitting due to narrow, institution-specific patterns (e.g., biases in imaging protocols) [114]. Federated learning addresses this by enabling collaborative model training without sharing raw data. Cross-device FL trains models locally on edge devices (e.g., wearables), exchanging encrypted updates [115], while cross-silo FL streamlines hospital collaborations: a central server distributes models to clinics, which train locally using

TABLE 9. Comparison of diabetic retinopathy datasets based on limitations, composition, data type, and accessibility.

Dataset	Limitations	Composition	Data Type
Messidor [81]	Imbalanced classes; lacks diversity; no lesion annotations	1,200 fundus images captured at 3 different resolutions; labeled with 4 DR severity grades	2D fundus images
Messidor-2 [109]	Small size; limited annotations	1,748 fundus images with DR severity labels; used for benchmarking detection algorithms	2D fundus images
EyePACS [86]	Variable image quality; noisy labels; preprocessing often required	88,702 images from diabetic screening programs; labeled across 5 DR severity levels	2D fundus images
IDRiD [110]	Small sample; limited real-world variation; includes rich lesion-level annotations	516 high-resolution images with pixel-wise labels for microaneurysms, hemorrhages, and more	2D fundus images
DRIVE [111]	Not designed for DR severity classification; focuses on vessel segmentation	40 fundus images with manual annotations for vessel segmentation tasks	2D fundus images
STARE [112]	Not DR-specific; general retinal diseases; limited dataset size	400 annotated images covering DR and other retinal diseases	2D fundus images
APTOS [86]	Variable image quality due to lighting/focus differences; requires normalization	35,126 images from blindness detection challenge; labeled with DR grades	2D fundus images
OCTA	Limited public availability; niche modality; not widely adopted in clinical practice	3D OCT angiography scans used for microvascular structure analysis in DR	3D OCT scans

frameworks like TensorFlow Federated and return aggregated updates [116]. FL categorizes based on data distribution and collaboration scenarios (Figure 15). In ophthalmology,

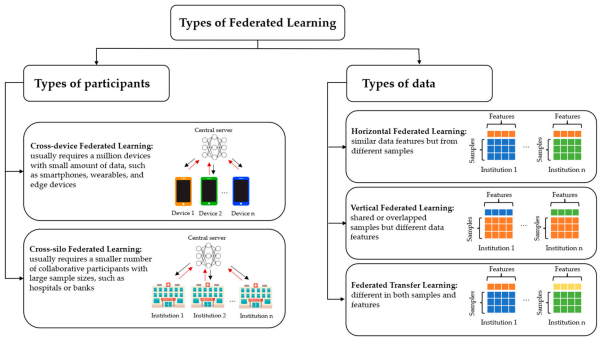


FIGURE 15. Types of federated learning [117].

FL advances retinal disease detection using modalities like OCT and OCTA. For diabetic retinopathy (DR), FL models trained on multi-institutional OCT/OCTA datasets matched institution-specific performance and surpassed external models, demonstrating cross-center viability [118]. Similarly, FL achieved microvasculature segmentation accuracy comparable to centralized training across four OCT devices, enhancing generalizability through diverse data [119]. For retinopathy of prematurity (ROP), FL harmonized subjective severity assessments across seven U.S. hospitals without expert consensus, achieving AUROC values (0.93–0.96) rivaling centralized training despite data variability [117]. Challenges persist, including small OCTA datasets limiting clinical relevance, OCTA's niche role compared to fundus photography, and unresolved risks like communication inefficiency or data bias [117], [118]. While FL reduces reliance on single-source data, hurdles like non-IID data distributions, security risks, and the non-explainable nature of the model—hindering bias assessment and fairness—remain critical. Addressing these requires integrating explainable AI (XAI) to enhance transparency, bridging the gap between decentralized learning and actionable clinical trust.

4) EXPLAINABLE AI (XAI)

Explainable AI refers to artificial intelligence designed to clarify its objectives, reasoning, and decision-making process in a manner that an ordinary person can comprehend. XAI enables human users to grasp the logic behind AI and machine learning algoithms, fostering greater confidence in their outputs. XAI is frequently associated with deep learning models and plays a significant role in the FAT (fairness,accountability, and transparency) principles of ML. It is valuable for organizations aiming to implement AI model responsibly. XAI assists developers in analyzing an AI system's behaviour, understanding how it reaches specific conclusions, and identifying potential concerns such as algorithmic biases. ML models are generally classified as *trasnparent(white box)* and *opaque(non-explainable)* systems. Transparent models greater clarity and produce results that are easier to interpret for both users and developers. Conversely, opaque models, such as neural networks, generate decisions that are difficult to decode- even for AI

specialist [120]. In the context of DR detection, XAI enhances the diagnostics process by making AI-driven analyses more interpretable. Techniques such as heatmaps and feature importance scores can highlight s specific retinal features that the model associates with the disease presence or progression. This level of explanantion not only aide clinicians in validating AI recommendations but also contributes to more personalized patient care. By understanding which features are deemed critical by the model, healthcare professionals can make more informed decisions regarding diagnosis and treatment strategies [121]. Drawing from the article's use of Layer-wise Relevance Propagation (LRP) with VGG16 to enhance transparency in skin disease diagnosis, similar XAI techniques can make deep learning models for DR more interpretable. By highlighting key features in retinal images that influence predictions, LRP can help clinicians trust and utilize AI systems effectively, improving diagnostic accuracy and patient care in medical settings. The study ''Explainable Artificial Intelligence Enabled TeleOpthalmology for Diabetic Retinnopathy Grading and classification'' demonstrates the pratical application of XAI in this doman. The researchers developed an XAI-enabled teleopthalmology model that not only grades and classifies DR but also provides visual explanations for its predictions. This approach improves the transparency of the model, allowing clinicians to see which areas of retinal images influenced AI decisions, thus building trust and facilitating better clinical decision-making [122]. The study utilized retinal fundus images for publicly available datasets. These images were subjected to preprocessing steps such as resizing, normalization, and augmentation to improve the robustness of the model. A convolutional neural network was designed to extract features from retinal images. The architecture included multiple convolutional layers, pooling layers and fully connected layers to effectively capture the hierarchical features relevant to DR classification. The model as trained using the preprocessed images, with a portion of the data reserved for validation to monitor performance and prevent overfitting. To make the model predictions interpretable, the study incorporated XAI methods such as Grad-CAM (Gradient-weighted Class Activation Mapping). This technique generated heatmaps highlighting regions of the input image that significantly influence the model's decision, allowing clinicians to understand the rationale behind each prediction. The developed model was integrated into teleophtalmology platform, enabling remote DR screening and diagnosis. The system allows healthcare providers to upload retinal images, recieves automated DR grading, and view the corresponding explanatory heatmaps [122]. The performance of the proposed model was assessed using several evaluation metrics: *1) Accuracy* The model achieved accuracy of 98.45% indicating the proportion of correctly classified instances among the total instances. *2) Sensitivity(Recall)* A sensitivity of 98.67% was reported, reflecting the model's ability to correctly identify positive cases of diabetic retinopathy. *3) Specificity* The specificity was 98.12%, demonstrating the model's capability to correctly identify negative cases, thereby reducing false positives. The model attained an F1-score of 98.56% which is the harmonic mean of precision and recall, providing a balance between the two metrics [122]. The integration of XAI into teleopthalmology system for DR grading and classification enhances the interpretability of AI-driven diagnoses. By providing visual explanations for its predictions, the system aims to improve clinicians trust and acceptance of AI tools in ophthalmic practice. The methodology demonstrates a significant step towards more transparent reliable AI applications in medical diagnostics.

TABLE 10. Comparative diagnostic performance of diabetic retinopathy detection approaches.

Approach	Typical Diagnostic Metrics	Strengths and Limitations
Traditional Methods	Accuracy:70–85%, Sensitivity: 65–80%, Specificity: 70–85%	Well-established and interpretable; expert-dependent and time-consuming
Machine Learning	AUC-ROC:0.80–0.90, Sensitivity: 75–85%	Automated feature extraction; requires handcrafted features
Deep Learning	Accuracy: 85–95%, F1-score: 80–92%, AUC-ROC: 0.90–0.98	End-to-end learning with high precision;non-explainable models nature and data-hungry
Federated Learning	Accuracy: 88–93%, Privacy compliance	Preserves data privacy; communication overhead
Explainable AI (XAI)	Accuracy: 85–92%, Improved interpretability	Enhances clinical trust; integration challenges

III. EVALUATION METRICS

Evaluating diabetic retinopathy detection methodologies requires a structured analysis of diagnostic performance, computational efficiency, and real-world applicability. Traditional techniques such as fundus photography and optical coherence tomography (OCT) are benchmarked based on sensitivity, specificity, and diagnostic accuracy. Nevertheless, AI methods like ML, DL, FL, and XAI go above basic accuracy when detecting diabetic retinopathy. They need additional checks, such as the AUC-ROC curve to see how well they split healthy from unhealthy cases, the F1-score to balance recall and precision, and computational complexity since they can be resource hogs. Federated learning keeps patient data private by training models locally, so we look at stuff like communication costs and how fast the model sharpens up. XAI, on the other hand, makes these tools easier to understand, with measures such as the accuracy of the saliency map that show what the model focuses on and how much doctors trust it. Blending all these factors gives a solid way to judge these systems, requiring them to be reliable and practical for real clinic use. Table 10 summarizes the typical diagnostic performance metrics reported across the reviewed studies for DR detection approaches. Traditional methods relying on manual or semi-automated analysis,show moderate accuracy and sensitivity, reflecting their dependence on expert interpretation and limited scalability. Machine learning improves these metrics by automating feature extraction, achieving AUC-ROC values typically between 0.80 and 0.90.Deep learning achieves high diagnostic accuracy (often>90%) and AUC-ROC upto

0.98 by learning hierarchical features and end-to-end, but its non-explainable nature limits clinical adoption. Federated learning ensures privacy with competitive acccuracy (88-93%) but adds communication challenges. Metrics summarize 116 studies using fundus and OCT imaging, reflecting AI's evolving role in DR detection.

TABLE 11. Diagnostic performance metrics.

Metric	Formula	Description
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Proportion of correctly predicted cases (both positive and negative).
Sensitivity (Recall)	$\frac{TP}{TP+FN}$	Measures how well the model identifies actual disease cases (true positives).
Specificity	$\frac{TN}{TN+FP}$	Measures how well the model identifies actual healthy cases (true negatives).

TP (True Positives) refers to correctly identified disease cases. TN (True Negatives) are correctly identified healthy cases. FP (False Positives) are healthy cases incorrectly labeled as diseased. FN (False Negatives) are disease cases missed by the model.

In Table 10, sensitivity, specificity, and accuracy were prioritized due to their high clinical relevance in diabetic retinopathy screening. Sensitivity ensures early detection of true DR cases, specificity helps avoid unnecessary treatments for healthy patients, and accuracy provides an overall evaluation of the model's performance. While F1-score and precision are crucial in imbalanced datasets, they were not included in Table 10 due to inconsistent reporting across many reviewed studies. However, where these metrics were available—such as in the XAI section—we have explicitly included and discussed them. This reflects our awareness of their importance, particularly in scenarios where class imbalance could affect evaluation.

A. CERTAINITY OF EVIDENCE

To check the robustness of evidence, we had made a GRADE summary for key results, including diagnostic accuracy,specificty and sensitivity. The table 12 presents a summary of the results along with an overall certainity rating. Each certainity rating tells our evaluation of factors such as risk of bias, inconsistency and imprecision accross the studies included. Detailed explanations for these ratings are provided in the table footnotes.

B. RANKED OPEN CHALLENGES IN AI-BASED DIABETIC RETINOPATHY SCREENING

To guide future research and implementation efforts, we summarize and prioritize the key open challenges in this domain:

1) **Dataset Bias and Lack of Diversity**: Many datasets rely on constrained datasets that lack population diversity, resulting in limited generalizability across real-world scenarios.

TABLE 12. Outcome, effect estimate, certainty rating, and explanation.

Outcome	Effect Estimate	Certainty Rating	Explanation (Footnotes)
Diagnostic Accuracy	92% (Range: 85–97%)	Moderate	Downgraded for risk of bias and variability across studies.
Sensitivity	90% (Range: 87–95%)	High	Consistent findings from large-scale studies.
Specificity	95% (Range: 90–98%)	Moderate	Downgraded due to imprecision in some included studies.

2) **Model Interpretability and Clinical Trust**: Deep learning systems are non-explainable in nature, which hinders clinicians ability to interpret or fully trsut their outputs.
3) **Data Privacy and Security Concerns**: Centralized data training can lead to privacy violations and legal concerns. Federated learning addresses these issues by keeping data local, yet it is hindered by implementation difficulties and resource demands.
4) **Infrastructure and Accessibility Barriers**: The implementation of AI-driven diagnostic systems remains challenging in economically disadvantaged areas due to prohibitive costs of hardware and operational infrastructure.
5) **Real-world Validation and Clinical Integration**: While AI models perform effectively in research trials, their real-world applicability is uncertain due to limited testing across varied clinical environments.

IV. DISCUSSION

Recent advancements in artificial intelligence have transformed diabetic retinopathy screening, enabling faster, more precise diagnoses and scalable solutions. Fundus imaging and optical coherence tomography are still commonly used diagnostic tools but come with certain limitations. Fundus cameras capture 2D retinal images, whereas OCT generates detailed 3D cross-sections of retinal layers. Despite their clinical utility, these methods demand costly equipment, specialized training, and infrastructure—challenges that limit accessibility in low-resource regions.

While AI-driven systems hold great potential, several obstacles impede their broad adoption. A key hurdle is the reliance on extensive annotated datasets for training. Many algorithms are trained on homogeneous datasets, leading to biases and reduced accuracy when applied to diverse patient demographics or imaging conditions. For instance, models trained on datasets lacking ethnic or socioeconomic diversity may underperform in underserved communities. Additionally, the opacity of deep learning architectures, such as convolutional neural networks, hinders clinical trust. Despite high diagnostic accuracy, their "non-explainable" nature complicates result interpretation. Emerging explainability tools—such as attention maps and saliency visualizations—

aim to clarify decision pathways, yet their integration into clinical workflows remains a challenge.

Data privacy and security further complicate AI deployment. Centralized training models necessitate aggregating sensitive patient data, raising ethical and compliance concerns. Federated learning, a decentralized approach where models train locally on institutional data, offers a privacy-preserving alternative. However, this method struggles with communication inefficiencies, uneven data quality across sites, and high computational costs. Resource constraints in rural or underfunded clinics exacerbate these issues, as AI systems often demand robust hardware and connectivity. Another gap lies in real-world validation. Most AI models are tested in controlled research settings using curated datasets, raising questions about their efficacy in dynamic clinical environments. To bridge this, future research must prioritize multi-institutional trials with diverse patient cohorts. Hybrid systems that merge fundus images, OCT scans, and electronic health records could enhance diagnostic robustness. Simultaneously, advancing explainable AI (XAI) frameworks and refining federated learning protocols will be crucial for fostering clinician trust, ensuring regulatory compliance, and enabling equitable global deployment of DR screening technologies.

V. CONCLUSION

AI-powered solutions are remodelling DR diagnostics, offering inflate accuracy, efficiency, and availability. The advancements from conventional ML techniques to DL-driven models has crucially improved feature extraction and categorization, surpassing the restriction of traditional screening techniques. While the evidence indicates high senstivity and resonable specificity,the moderate certainity rating for diagnostics accuracy highlights the need for high-quality research to address potential biases and imprecision. FL and XAI have further present to addressing bother related to data privacy, clarity, and model explicability. However, despite these improvements, challenges such as restricted dataset variety, high computational requests, and the need for factual clinical authentication persist.

To ensure general clinical assumption, future exploration must focus on evolve multi-modal AI foundation that consolidate fundus imaging, OCT, and impersonal data for a more complete diagnosis. Enhancing XAI methods will be essential in stimulate assurance among healthcare professionals, verify AI-driven models position with clinical decision-making procedures. Additionally, rank the deployment of AI-based screening machines in resource-limited settings can ease of early detection and intervention, eventually reducing the overall burden of diabetes-related eyesight impairment. By addressing these provocations, AI-driven DR detection can conversion from experimental models to fully combined clinical solutions, restructure ophthalmic disease control and patient awareness.

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